VARBARO-TYPE CONNECTEDNESS: THE COMPLEMENT OF A SMALL CLOSED SUBSET IN V(I) STAYS CONNECTED UNDER A COHOMOLOGICAL-DIMENSION BOUND

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4585-4602 of sources/stacks-algebraization/pinned-original-source.tex; proof span is lines 4604-4662, the complete author proof (57 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool algebraization wave36 (al02 bib-salvage batch). Spans computed from \label (16.5/16.6).

Editorial note. External \cite keys salvaged with card-local bibliography.tex entries copied verbatim from the snapshot my.bib (precedent S3-morph-blow-up-ample-family); URL/ISSN/EPRINT fields dropped as the preamble has no url package.

Editorial note. statement contains reference environment with real content -> patch-sd-envs.tex applied (sd02 lesson).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; in-chapter lemmas cited by name become stub prereqs; external cites resolved via the card-local bibliography.tex fragment.

Source reference. Original lemma 090T: sources/stacks-algebraization/pinned-original-source.tex, source lines 4662–4662. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BFR: sources/stacks-algebraization/pinned-original-source.tex, source lines 4662–4662. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DXR: sources/stacks-algebraization/pinned-original-source.tex, source lines 3849–3866. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DX9: sources/stacks-algebraization/pinned-original-source.tex, source lines 4662–4662. Cited by the retained statement or

Date: stacks-project:algebraization.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DXF: sources/stacks-algebraization/pinned-original-source.tex, source lines 4662–4662. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ECP: sources/stacks-algebraization/pinned-original-source.tex, source lines 4662–4662. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02J6: sources/stacks-algebraization/pinned-original-source.tex, source lines 4662–4662. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. [1, Theorem 1.6] *Let $(A, \mathfrak{m})$ be a Noetherian complete local ring. Let I be a proper ideal of A. Set $X = \operatorname{Spec}(A)$ and $Y = V(I)$. Denote*

1. *d the minimal dimension of an irreducible component of X, and*
2. *c the minimal dimension of a closed subset $Z \subset X$ such that $X \setminus Z$ is disconnected.*

Then for $Z \subset Y$ closed we have $Y \setminus Z$ is connected if $\dim(Z) < \min(c, d-1) - cd(A, I)$. In particular, the punctured spectrum of A/I is connected if $cd(A, I) < \min(c, d-1)$.

Proof. Let us first prove the final assertion. As a first case, if the punctured spectrum of A/I is empty, then Local Cohomology, Lemma 0DXF shows every irreducible component of X has dimension $\leq \operatorname{cd}(A, I)$ and we get $\min(c, d-1) - \operatorname{cd}(A, I) < 0$ which implies the lemma holds in this case. Thus we may assume $U \cap Y$ is nonempty where $U = X \setminus \{\mathfrak{m}\}$ is the punctured spectrum of A. We may replace A by its reduction. Observe that A has a dualizing complex (Dualizing Complexes, Lemma 0BFR) and that A is complete with respect to I (Algebra, Lemma 090T). If we assume $d - 1 > \operatorname{cd}(A, I)$, then we may apply Lemma 0DXR to see that

$$\operatorname{colim} H^0(V, \mathcal{O}_V) \longrightarrow \lim H^0(U, \mathcal{O}_U/I^n\mathcal{O}_U)$$

is an isomorphism where the colimit is over opens $V \subset U$ containing $U \cap Y$. If $U \cap Y$ is disconnected, then its nth infinitesimal neighbourhood in U is disconnected for all n and we find the right hand side has a nontrivial idempotent (here we use that $U \cap Y$ is nonempty). Thus we can find a V which is disconnected. Set $Z = X \setminus V$. By Local Cohomology, Lemma 0DXF we see that every irreducible component of Z has dimension $\leq \operatorname{cd}(A, I)$. Hence $c \leq \operatorname{cd}(A, I)$ and this indeed proves the final statement.

We can deduce the statement of the lemma from what we just proved as follows. Suppose that $Z \subset Y$ closed and $Y \setminus Z$ is disconnected and $\dim(Z) = e$. Recall that a connected space is nonempty by convention. Hence we conclude either (a) $Y = Z$ or (b) $Y \setminus Z = W_1 \amalg W_2$ with W_i nonempty, open, and closed in $Y \setminus Z$. In case (b) we may pick points $w_i \in W_i$ which are closed in U, see Morphisms, Lemma 02J6. Then we can find $f_1, \ldots, f_e \in \mathfrak{m}$ such that $V(f_1, \ldots, f_e) \cap Z = \{\mathfrak{m}\}$ and in case (b) we may assume $w_i \in V(f_1, \ldots, f_e)$. Namely, we can inductively using prime avoidance choose f_i such that $\dim V(f_1, \ldots, f_i) \cap Z = e - i$ and such that in case (b) we have $w_1, w_2 \in V(f_i)$. It follows that the punctured spectrum of $A/I + (f_1, \ldots, f_e)$ is

disconnected (small detail omitted). Since $\mathrm{cd}(A, I + (f_1, \ldots, f_e)) \leq \mathrm{cd}(A, I) + e$ by Local Cohomology, Lemmas 0ECP and 0DX9 we conclude that

$$\mathrm{cd}(A, I) + e \geq \min(c, d - 1)$$

by the first part of the proof. This implies $e \geq \min(c, d-1) - \mathrm{cd}(A, I)$ which is what we had to show. □

References

[1] Varbaro, Matteo. *Gröbner deformations, connectedness and cohomological dimension.* J. Algebra 322 (7), 2492–2507 (2009).